

CM 6004 Applied Digital Innovations



Chutisant Kerdvibulvech, Ph.D.
Associate Professor

รศ.ดร.ชุตินันต์ เกตวิบูลย์เวช

L1

INSTRUCTOR

Assoc. Prof. Dr. Chutisant Kerdvibulvech

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(Please Email/Facebook me first to make sure that I will be there.)

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Course Syllabus and Lectures

For protecting our environment, all course syllabus and lectures can be downloaded online via the following links.

- Course Syllabus
 - <http://gscm.nida.ac.th/~chutisant/CM6004/CS-CM6004.pdf>
- Lectures
 - <http://gscm.nida.ac.th/~chutisant/CM6004/L1-CM6004.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM6004/L2-CM6004.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM6004/L3-CM6004.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM6004/L4-CM6004.pdf>

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Evaluation (50%)

- Assignments, Participation and Attendance
10 %
- Project
40 %
(Technique-based, Creativity-based, and Research-based)

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Class Schedule

นวัตกรรมของสื่อดิจิทัลและเทคโนโลยีสารสนเทศในปัจจุบัน	๓	บรรยายเชิงอภิปราย
สื่อดิจิทัลในรูปแบบเทคโนโลยีใหม่: การรับรู้ทางสายตาและภาพลวงตาสำหรับสื่อดิจิทัล	๓	บรรยายเชิงอภิปราย
สื่อดิจิทัลในรูปแบบเทคโนโลยีใหม่: การรับรู้ทางสายตาและภาพลวงตาสำหรับสื่อดิจิทัล (ต่อ)	๓	บรรยายเชิงอภิปราย
งานวิจัยของสื่อดิจิทัลและเทคโนโลยีสารสนเทศในปัจจุบัน	๓	บรรยายเชิงอภิปราย
ปัญหาของสื่อดิจิทัลในแง่เทคโนโลยีสารสนเทศในปัจจุบัน	๓	บรรยายเชิงอภิปราย
อนาคตของความเกี่ยวข้องกันของสื่อดิจิทัลและเทคโนโลยีสารสนเทศ	๓	บรรยายเชิงอภิปราย
ทิศทางของการบูรณาการของสื่อดิจิทัลและเทคโนโลยีสารสนเทศ	๓	บรรยายเชิงอภิปราย

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Textbooks

□ Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, ISBN 9786167687216

□ ชุดลิ้นต์ เกิดวิบูลย์เวช. (2560) (2559) สื่อดิจิทัลใหม่..สื่อแห่งอนาคต. คณะวิทยาศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์.

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Academic Paper (Scopus)

- C. Kerdvibulvech, "A Methodology for Hand and Fingers Motion Analysis Using Adaptive Probabilistic Models," EURASIP Journal on Embedded Systems, 9 pages, No. 18, Springer Publisher, 2014.

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Websites and Links

- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/bookseries/558>
- <http://www.dailynews.co.th/>
- <http://www.ecommerce-magazine.com/>
- <http://www.manager.co.th>

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Overview

- ▣ Background
- ▣ From Digital Media to New Digital Media
- ▣ 3D Display / 3D Television
- ▣ Augmented Reality

Assoc. Prof. Dr. Chutsant Kerdvibulvech

Academic Work Experience

- ▣ Invited Lecturer and Research Committee in many universities in Thailand, such as
 - Chulalongkorn University
 - Sripatum University
 - Panyapiwat Institute of Management
 - Mahidol University
 - Kasetsart University
- ▣ Technical Committees for many academic international conferences and journals (Scopus and Scimago)

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Myself

- ▣ Received PhD in Computer Engineering from Keio University, Japan (2009)
- ▣ Assistant Professor (2015-2018), Associate Professor (2018-Now) at GSCM, NIDA
- ▣ Head of Department of Information and Communication Technology, Rangsit University (2009-2014)

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Media Work Experience

Newspapers

- ▣ IT Columnist for Dailynews newspaper, every Wednesday (page 23)



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FROM DIGITAL MEDIA TO NEW DIGITAL MEDIA

Digital Media

Digital media are any media that are encoded in a machine-readable format. [Technology Brief. University of Guelph.]

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



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Important Role of Digital Media

Online video has become ubiquitous, digital media consumption has become the norm.

Facts about Digital Media

1. Viewers are heavily engaged with digital media – less so with linear television and newspaper.
2. Television and newspaper still matter, but it can no longer succeed as a stand alone platform.

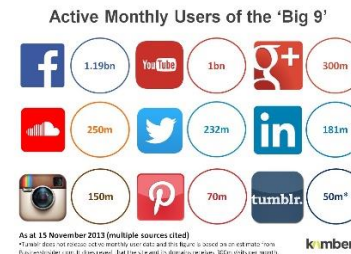


::Source::
Digital Media Trends and the Future of
Television, Dr. Seth Geiger,
President SmithGeiger LLC (CA, USA)
Assoc. Prof. Dr. Chutisant Kerdvibulvech

<https://www.linkedin.com/>

Trends for Digital Media for 2014

Active monthly users of the 'big 9'



::Source::
Kamber (Sydney,
Australia)

Trends for Digital Media for 2016



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<https://www.facebook.com/zuck>

Mobile Phone As Home Computer

- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential

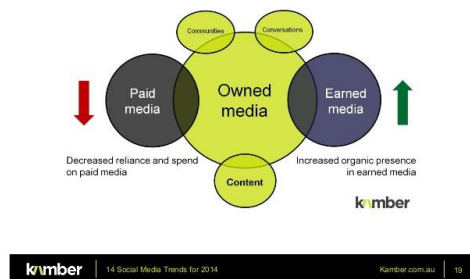


voices.suntimes.com

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Paid Media / Earned Media

The content marketing argument



Source:
 Kamber (Sydney, Australia)

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Creative-Digital Economy

- Placement: Its where young (and old) people are
- Price to invest: Minimal
- But..
 - Easy to build innovatively
 - Create HUGE profits



<http://www.dailynews.co.th/>

Future of Digital Media?



- Big Data / SEO (Search Engine Optimization) / SEM (Search Engine Marketing)
- AI (Artificial Intelligence) / Deep Learning
- Reaching-Out-of-Computer
 - ▣ 3D-Display / 3D Television
 - ▣ MR (Mixed Reality) / AR (Augmented Reality) / VR (Virtual Reality)
 - ▣ New Digital Media

www.3dsg.com.au

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3D DISPLAY / 3D TELEVISION

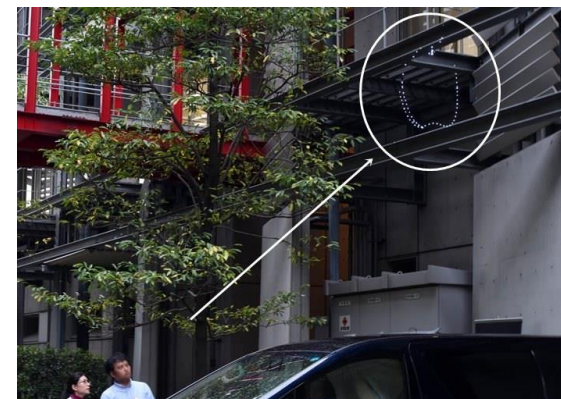
3D Display / 3D Television



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3D Display / 3D Television

A free-floating image created by firing lasers into thin air

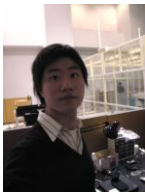


<http://www.afp.com/>

3D Display / 3D Television

2014 October

- 2008-Current
- Hideki Kimura
- National Museum of Emerging Science and Innovation (Miraikan), Tokyo



2009



<http://www.afp.com/>

3D Display / 3D Television



<http://www.afp.com/>

3D Display / 3D Television

- Miraikan is located in Odaiba, Tokyo



www.whitehouse.gov

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www.japan-guide.com

NEW VIEWPOINT ANALYSIS

Prof. Takeo Kanade,
Carnegie Mellon University (CMU), United States

Super Bowl

- Football fans treated to a unique new view of the action during playbacks
- CBS Television used a new technology
- Carnegie Mellon University computer vision expert, Prof. Takeo Kanade

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Super Bowl

- “Eye Vision”
- Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



Eye Vision from Prof. Takeo Kanade, CMU

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The Matrix



www.imdb.com

- “Virtualized Reality”
- People will be able to customize the perspective from which they watch, for example, from that of a particular object.



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New Viewpoint Analysis

AUGMENTED REALITY

What is Augmented Reality (AR)?



- A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- Generates a composite view for the user.

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What is the Goal of AR?

- To enhance a person's performance and perception of the world
- But, what is the ultimate goal?



["Augmented reality in urban places: contested content and the duplicity of code."](#)

Transactions of the Institute of British Geographers, 2012

The Ultimate Goal of AR

- Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.



AR in Commerce Geek.com

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Augmented Reality vs. Virtual Reality

Augmented Reality

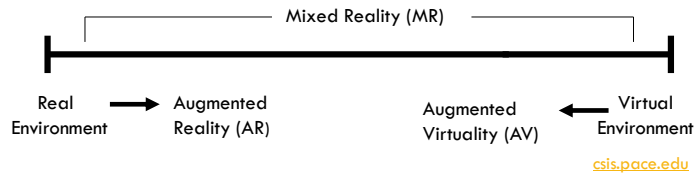
- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

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Miligram's Reality-Virtuality Continuum



Miligram coined the term "Augmented Virtuality" to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

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Combining the Real and Virtual Worlds

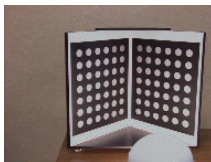
We need:

- Precise models
- Locations and optical properties of the viewer (or camera) and the display
- Calibration of all devices
- To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

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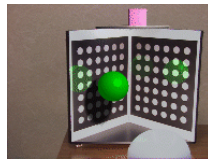
Combining the Real and Virtual Worlds (cont)

- Register models of all 3D objects of interest with their counterparts in the scene
- Track the objects over time when the user moves and interacts with the scene



real world

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real world with virtual objects and inter-reflections and virtual shading

Mixed Reality

- Human pacman: A mobile wide-area entertainment system based on physical, social, and ubiquitous computing

Journal: Personal and Ubiquitous Computing



2004
Dr. Adrian Cheok,
National University of Singapore (NUS)
at Singapore

Broadcast in Discovery Channel

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This image depicts an artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



www.sonycs.jp

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Realistic Merging

Requires:

- Objects to behave in physically plausible manners when manipulated
- Occlusion
- Collision detection
- Shadows

****All of this requires a very detailed description of the physical scene**

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Research Activities

- Develop methods to register the two distinct sets of images and keep them registered in real-time
 - ▣ New work in this area has started to use Computer Vision (CV) techniques
- Develop new display technologies for merging the two images

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Performance Issues

Augmented Reality systems are expected:

- To run in real-time so that the user can move around freely in the environment
- Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- Update rate for generating the augmenting image
- Accuracy of the registration of the real and virtual image

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Failures in Registration

Failures in registration due to:

- ❑ Noise

- Position and pose of camera with respect to the real scene
- Fluctuations of values while the system is running

- ❑ Time delays

- In calculating the camera position
- In calculating the correct alignment of the graphics camera

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Applications

- ❑ Medical
- ❑ Military Training
- ❑ Hazard Detection
- ❑ Engineering Design
- ❑ Robotics and Telerobotics
- ❑ Training
- ❑ Sports
- ❑ Tourism
- ❑ Video conferencing

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Applications (cont.)

- ❑ Entertainment, Computer Games
- ❑ Manufacturing, Maintenance, and Repair
- ❑ Consumer Design
- ❑ Audio
- ❑ Education
- ❑ Architecture
- ❑ Human welfare

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Interactive Digital Multimedia Applications

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AUGMENTED REALITY APPLICATIONS

SIXTH SENSE

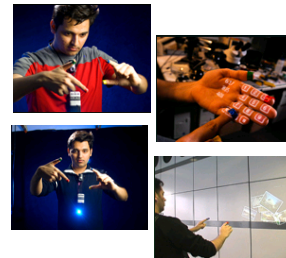
Sixth Sense

- A wearable gestural interface that augments the physical world around us with digital information
- Let us use natural hand gestures to interact with that information
- The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

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Sixth Sense in 21st Century

TED
TALKS



Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED

2



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Detail of Sixth Sense

- Sixth Sense allows people to use the internet without a screen or a keyboard
- It can turn any surface into a touch screen
- It is made with store-bought components
- *"The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques"..from MIT Media Lab*

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Detail of Sixth Sense

- The device recognizes items in stores, and then gathers and projects information about products to the user.
- It also is able to provide quick signals to let users know what the best products are based on their needs."
- Recognize articles in newspapers, retrieve the latest related stories or video from the Internet and then display them on pages for the user
- Say goodbye to your encyclopedia

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Cost of Sixth Sense

- Only \$350 dollars, around 12,000 baht
- From an educator's stand point, this is minimum given the amount of money spent on textbooks, computer labs and LCD projectors.
- For educator's, Sixth Sense would be a way to put technology (literally) into each student's hands, because of its size and its cost.

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AUGMENTED REALITY
GUITAR

Augmented Reality Guitar



Dr. Chutisant Kerdvibulvech,
Keio University, Japan

Broadcast in Discovery Channel



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Thank you

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